# Best Practices for Biomedical Research Data Management

## Module 8: Curation & Preservation for Data Digital Content Management: MIT Example

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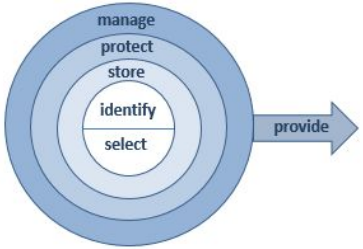


This video will provide an example of collaboration across an organization to manage digital content. This example of the 'Digital Content Management Initiatives' at MIT highlights the different components and departments involved in the workflow for reviewing digital content for curation and preservation.

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# DIGITAL CONTENT MANAGEMENT INITIATIVE AT MIT LIBRARIES

# Digital Content Management Levels

**Intellectual** – collections and acquisition strategy

**Workflow** – coordinated  life cycle management (DCM)

**Preservation** – strategies, formats, media (CPS)

**Archival Storage** – long-term management (CPS + IT)

**IT** – technical environment (e.g., systems, software)

# Life Cycle Roles

- **Producer /Creator** – local and beyond
- **Curator** – overall, content-specific, each stage
- **Archivist** – big A and little a
- **Manager** – at any level, points in time
- **Funder** – local and beyond
- **Advocate** – anyone
- **Consumer** – current and future

# DCM Activities

1. **Coordination** – life cycle group to raise awareness about life cycle management and oversee #2 and #3
2. **Digital content reviews** – overview of digital content types with life cycle impact assessments
3. **Digital content projects** – established life cycle experiments to respond to ongoing needs

# Digital Content Types

- Architectural and design documentation
- Audio
- Digital images (scanned)
- E-books
- Faculty archives
- Geospatial information
- **Research data**
- Theses
- Video
- Web content

Digital Object = metadata, content, packaging

# Digital Content Review

- Structured process to complete a gap analysis for majority of digital content types
- Uses a template with questions pertaining to life cycle stages including:
  - *selecting*, *acquiring*, *processing*, *storing*, *preserving*, *disseminating*, and *managing rights*
- Documents current status of workflow and practice for each digital content type
- Identifies implications for MIT Libraries of managing digital content types across generations of technology

# Purpose of DCR

- Identify short-term approach (approx. 3 years)
- Develop policies and procedures informed by community standards and practice
- Enable action while practice evolves for new/changing digital content types
- Provide baseline for updating and revising recommendations
- Provide model and examples for community

# Digital Content Review Results

Two parts: overviews and life cycle assessments

Overviews:

- Identify content with common characteristics ("buckets")
- Provide a concise view of each digital content type

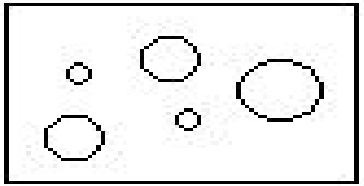
Life cycle assessments:

- Produce one or more concise reports per content type
- Highlight areas for improvement

Cumulatively, digital content reviews help to:

- Address content-specific and general requirements
- Prioritize digital content management improvements

# Identifying “Buckets”



A bucket contains digital objects within a content type that have similar characteristics

Buckets vary by digital content type

Factors that determine buckets include:

things that affect workflow, for example:

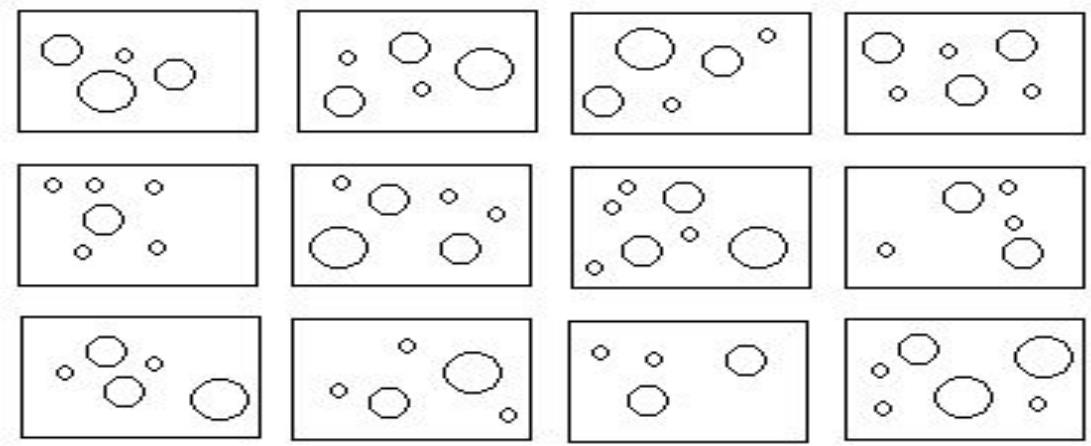
- where the content comes from (who provides it, source)
- what happens to it (creation, processing)

things that affect discovery and delivery, for example:

- how users find content
- how users get and use content

# DCR Overviews

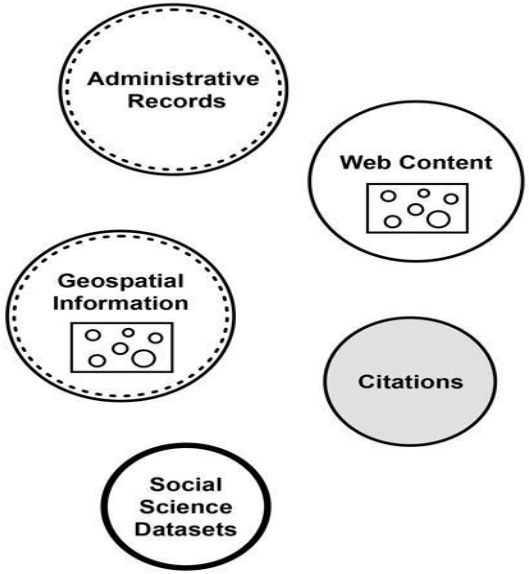
one overview per digital content type
cumulative overviews = landscape of digital content



Circles = birds eye views of buckets for each content type

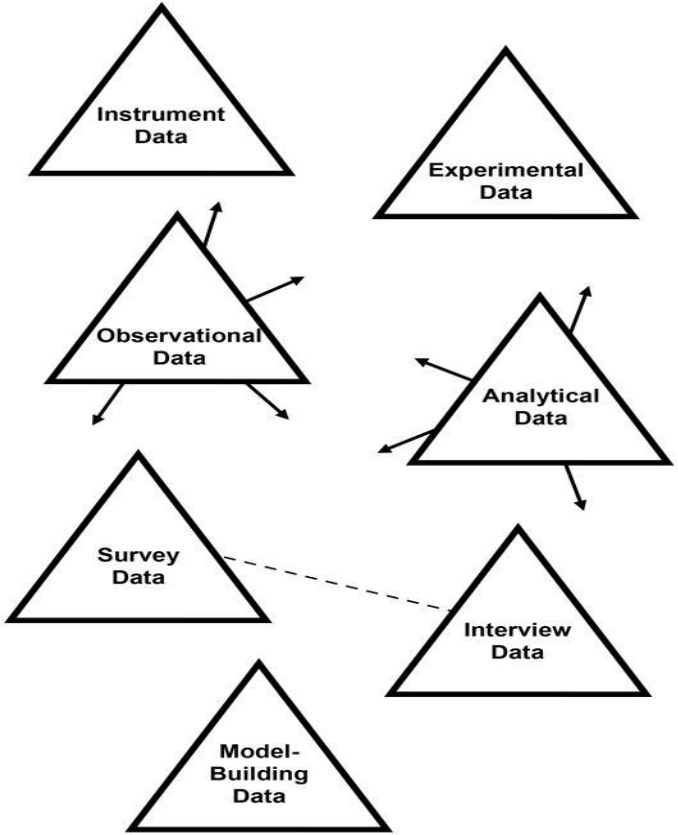
# Research Data

## Possible Sources for Research

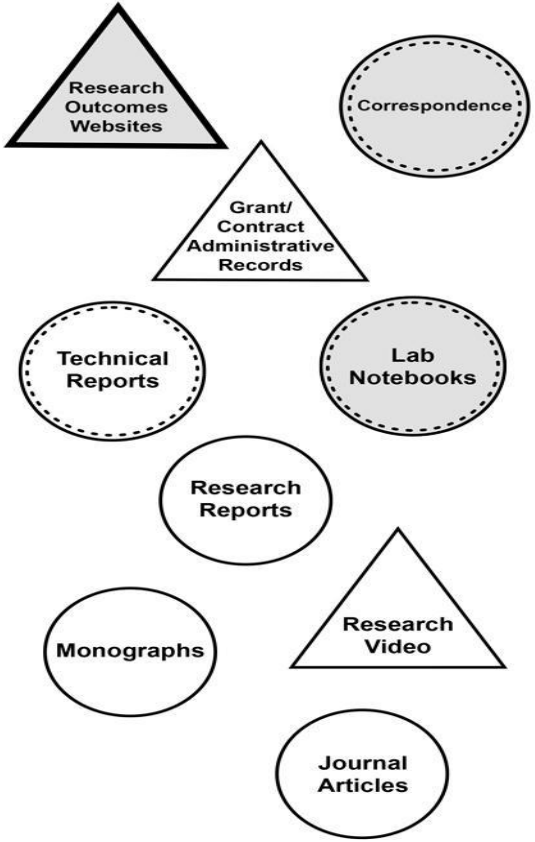



Thick outline = bucket is in its primary content type
Thin outline = bucket is not in its primary content type
Light gray bucket = content for which some amount is currently managed by the Libraries and that we expect to acquire more of within 3-5 years
Dotted plus solid outline = content is both digitized and received (born or acquired) digitally
Triangle = MIT content that is monitored but not managed by the Libraries; may or may not be acquired in the future
Small swiss cheese diagram = complex bucket that contains other buckets; may refer to an entire content type

## Research Data Produced at MIT




## Research-Related Content



# DCR Report Sections

- Collection development
- Rights Management Assessment
- Acquisition / Accessioning
- Ingest / Processing
- Preservation Planning
- Archival Storage
- Management / Roles
- Long-term Access / Dissemination